

Classical Threshold Generation for Oblivious Trees

Empirical study on the OBT pipeline — results & inferences

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Variational Update — April 2026

Building on the **Quantum Oblivious Trees** project, we paused the quantum direction to first *understand the classical threshold-generation landscape*.

The leaf-free, threshold-learning oblivious tree is novel even without quantum circuits — it deserves a thorough classical study before layering quantum methods on top.

Why this matters. The quantum circuit is just another *indirect* parameterization of the thresholds. If indirect parameterization gives no advantage classically, the rationale for the quantum comparison must be re-stated.

Architecture recap

- 1 A **threshold generator** outputs d thresholds $t = \{t_1, \dots, t_d\}$ (one per tree level).
- 2 Thresholds feed a **differentiable oblivious tree** $\rightarrow$ soft predictions $\hat{y}$ via sigmoid soft-binning + temperature annealing.
- 3 **Leaves are computed empirically** (responsibility-weighted majority vote) — not learned.
- 4 BCE / CE loss $\mathcal{L}(y, \hat{y}) \rightarrow$ gradients flow back to the threshold generator.
- 5 Generator parameters update; loop repeats. Sigmoid maps thresholds to $[0, 1]$.

The generator takes a *dummy input* — no data encoding.

The two approaches under comparison

	FFNN generator “CLASSICO FFNN”	Sampler “CLASSICO SAMPLER”
Threshold	$t_i = \sigma(W \cdot x + b_i)$	$t_i = \sigma(\theta_i), \theta_i \in \mathbb{R}$
Parameterization	Indirect (network)	Direct (d learnable scalars)
Hypothesis	more weights $\Rightarrow$ more expressive	minimal baseline
# parameters	$2d$ (0-layer) up to $\sim hd$ (1 hidden)	d

Question: does the indirect parameterization buy us anything?

Hypotheses we set out to test (Meeting 1)

- Q1. Is the FFNN necessary, or is direct optimization enough?
- Q2. Does adding hidden layers / neurons improve threshold quality?
- Q3. Convergence dynamics — speed vs. final accuracy tradeoff?
- Q4. Optimizer / learning-rate sensitivity?
- Q5. How does the differentiable OBT compare to a standard sklearn DT?

Experimental design — 4 studies

Study	Question	Sweep
1	FFNN-0layer vs Sampler	22 datasets $\times$ {FFNN-bias, FFNN-nobias, Sampler} $\times$ depths
2	Effect of hidden layers	22 datasets $\times$ $\{h = 4, 8, 16\}$ $\times$ depths
3	Optimizer / LR sensitivity	{Adam, SGD} $\times$ {0.001, 0.005, 0.01, 0.05, 0.1}
4	sklearn DT baseline	22 datasets, depth-matched DT

- **Datasets:** 22 tabular classification datasets.
- **Metrics:** test accuracy, test CE, validation curves.
- **Multiple seeds** per config; mean $\pm$ std reported.
- **Total:** $171 + 171 + 1086 + 57 = \mathbf{1485}$ runs.

Headline numbers

Study	Approach	Test acc	# params (avg)
1	FFNN-0layer (bias)	0.7046	14.5
1	FFNN-0layer (no bias)	0.7043	14.5
1	Sampler	0.7054	7.3
2	FFNN-1h, $h = 4$	0.7053	44
2	FFNN-1h, $h = 8$	0.7046	81
2	FFNN-1h, $h = 16$	0.7047	155
3	best (Adam, LR=0.01)	0.7020	—
4	sklearn DecisionTree	0.7129	—

All classical OBT variants are within ~ 0.001 of each other. DT is ~ 0.8 **pp** higher.

- Sampler and FFNN-0layer-no-bias are *mathematically* near-equivalent given a 1-D dummy input — $\sigma(w \cdot 1)$ is a reparameterized scalar.
- **Per-dataset accuracies are identical to 4 decimals on most datasets.**
- Sampler does this with **half the parameters** and one extra “win” overall.

Bias term effect: mean +0.002 (median +0.001), but ± 4 pp swings per dataset.

- Helps decisively on led7 (+0.11), car (+0.05).
- Hurts on australian (−0.08), ecoli (−0.06), heart (−0.05).
- Net: a wash, not a real effect.

Take-away

Indirect FFNN parameterization buys **nothing** over a direct learnable parameter. The “more weights $\Rightarrow$ more expressive thresholds” hypothesis **does not hold**.

Config	# params	Mean test acc
0-layer	14	0.7046
1 hidden, $h = 4$	44	0.7053
1 hidden, $h = 8$	81	0.7046
1 hidden, $h = 16$	155	0.7047

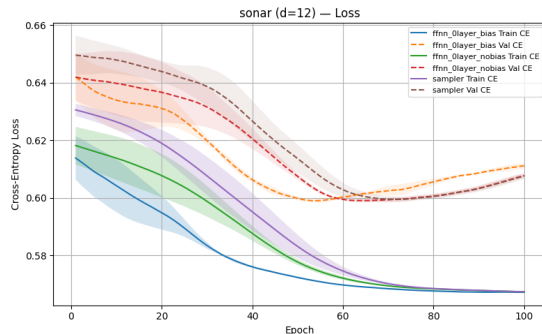
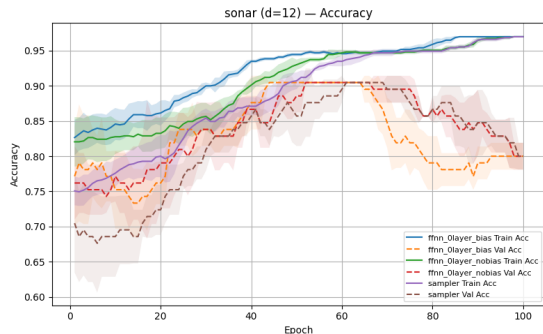
Going from **14** $\rightarrow$ **155** parameters yields a delta of $+0.0001$ in mean test accuracy. The 18-vs-4 win-count for $h = 16$ is rounding noise.

Take-away

The expressiveness bottleneck is **not** in the threshold generator. Whatever blocks further accuracy lives in the **differentiable oblivious tree itself** — soft binning, leaf computation, temperature schedule — not in how thresholds are produced.

Inference 3 — Convergence dynamics

(Q3)



sonar, depth = 12. Final accuracy is **tied**, but the dynamics differ.

- **FFNN-bias** starts higher, reaches its plateau **~30 epochs faster**.
- **Sampler** trains slowest, ends at the same point.
- FFNN validation **oscillates and degrades** after ~60 epochs (overfitting visible).
- Sampler's validation is **smoother** and more stable.

Same destination, different path: FFNN is faster but more variance / overfitting; Sampler is slower but better-behaved.

	best LR	best acc	spread (max – min)
FFNN, Adam	0.01	0.7020	0.013
FFNN, SGD	0.10	0.6760	0.022
Sampler, Adam	0.01	0.7019	0.023
Sampler, SGD	0.10	0.6664	0.018

- **Adam strictly dominates SGD** by ~ 3 pp regardless of approach. SGD never catches up, even at LR=0.1.
- **Sampler is $\sim 2\times$ more LR-sensitive** under Adam — drops to 0.679 at LR=0.001 while FFNN holds 0.689.
- **Adam @ LR=0.01** is the safe default for both.

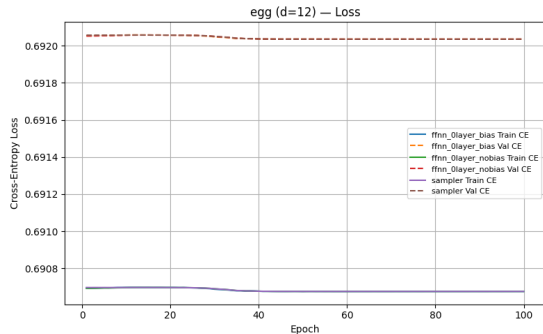
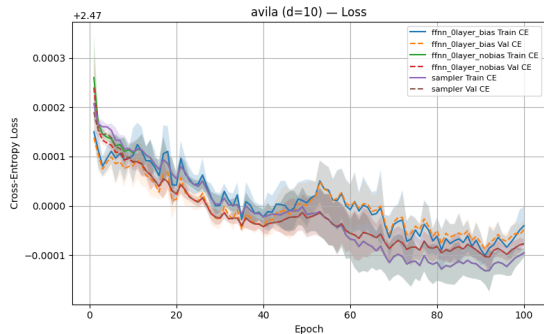
Per-dataset head-to-head (best Study-1 OBT vs sklearn DT, averaged over depths):

- **OBT wins 14/22**, DT wins 8/22. Mean diff +0.012 in favor of OBT.
- But DT *crushes* OBT on a **specific cluster of datasets**:

Dataset	OBT – DT	Notes
avila	−0.26	OBT stuck at 0.411 ($\approx$ majority class) at every depth
egg	−0.19	OBT stuck at 0.567 at every depth
drybean	−0.09	OBT improves with depth, DT just better
car	−0.08	
glass	−0.06	

OBT crushes DT on sonar (+0.18), australian (+0.14), lymph (+0.14), yeast (+0.11), pima (+0.10), heart (+0.09), iris (+0.08), fico (+0.06).

Inference 5 — what avila and egg look like

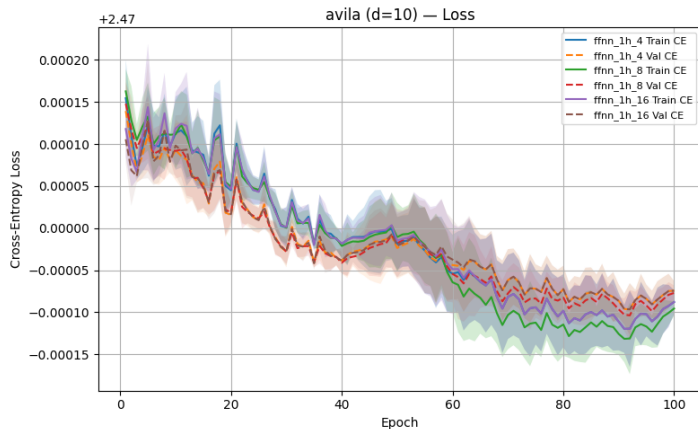


CE loss barely moves on avila ($2.4707 \rightarrow 2.4699$ over 100 epochs) and egg — for every classical OBT variant: FFNN-bias, FFNN-nobias, Sampler, $h = 4/8/16$.

Pattern

OBT wins on **binary / well-separated** datasets, loses on **multi-class** problems where greedy splitting matters. The collapse to majority-class is a **soft-binning landscape failure**, not a threshold-generator failure.

Hidden layers don't rescue avila either



Study 2, avila, depth = 10. $h = 4$, $h = 8$, $h = 16$ all flatline at the same loss — confirms the bottleneck is downstream of the threshold generator.

What this means for the project

- 1 **For the classical study.** Keep the cheapest baseline: **Sampler**. FFNN adds parameters and overfitting risk for **zero** accuracy gain. Hidden layers can be dropped from future grids.
- 2 **For the quantum direction.** The “indirect parameterization buys expressiveness” rationale that motivated the quantum comparison **didn't survive the classical ablation**. A quantum threshold generator must demonstrate something *other than* expressive thresholds — e.g.:
 - better optimization landscape,
 - robustness on hard datasets (*avila*, *egg*),
 - different inductive bias.
- 3 **The most interesting open problem** is *not* FFNN-vs-Sampler-vs-Quantum. It is:
Why does the soft-OBT collapse to majority-class on avila and egg regardless of who generates the thresholds?

Worth investigating before pouring more cycles into threshold generators.

Why is this the most interesting question?

- 1 **It is the actual bottleneck.** Studies 1–2 ruled out the threshold generator (Sampler $\approx$ FFNN-0layer $\approx$ FFNN-1h within ~ 0.001); Study 3 ruled out optimizer / LR. All those interventions live *upstream* of where *avila* / *egg* fail $\Rightarrow$ the failure must live in the soft-OBT layer itself.
- 2 **It re-frames the quantum question** — the whole point of the broader project. The classical ablation killed the “expressive thresholds” rationale. If a quantum generator is to add value, this finding tells us *where the room actually is*: the optimization landscape / inductive bias of the OBT layer, not threshold expressiveness.
- 3 **Sharp, falsifiable empirical signature.** On *avila*, CE loss moves $2.4707 \rightarrow 2.4699$ over 100 epochs, across every generator variant and every depth — while *sonar* / *iris* train fine. Two clean regimes from the same architecture $\Rightarrow$ tractable to diagnose (loss-landscape probe, gradient-norm analysis, temperature ablation).
- 4 **A class of failure, not a quirk.** Losing cluster (*avila*, *egg*, *drybean*, *car*, *glass*) is *multi-class*; winning cluster is binary / well-separated. The pattern likely generalizes to other soft-decision models (NODE, soft trees).
- 5 **Opportunity cost.** Generator-vs-generator competition is now bounded above by ~ 0.001 in mean accuracy. Fixing *avila* alone is bounded above by $+0.26$ **on a single dataset**. That is where the cycles pay off.

Proposed next steps

- **Diagnose the soft-binning failure** on avila / egg:
 - loss-landscape inspection at the OBT layer (not the generator),
 - temperature-schedule ablation; revisit annealing,
 - leaf-computation alternatives (responsibility-weighting variants).
- **Drop FFNN from the grid**; promote Sampler as the classical reference.
- **Re-frame the quantum question**: target *optimization landscape* or *inductive bias*, not threshold expressiveness.
- **Multi-class focus**: a sub-grid of multi-class datasets to track the OBT failure mode under interventions.